Decoding

NLP: Fall 2025

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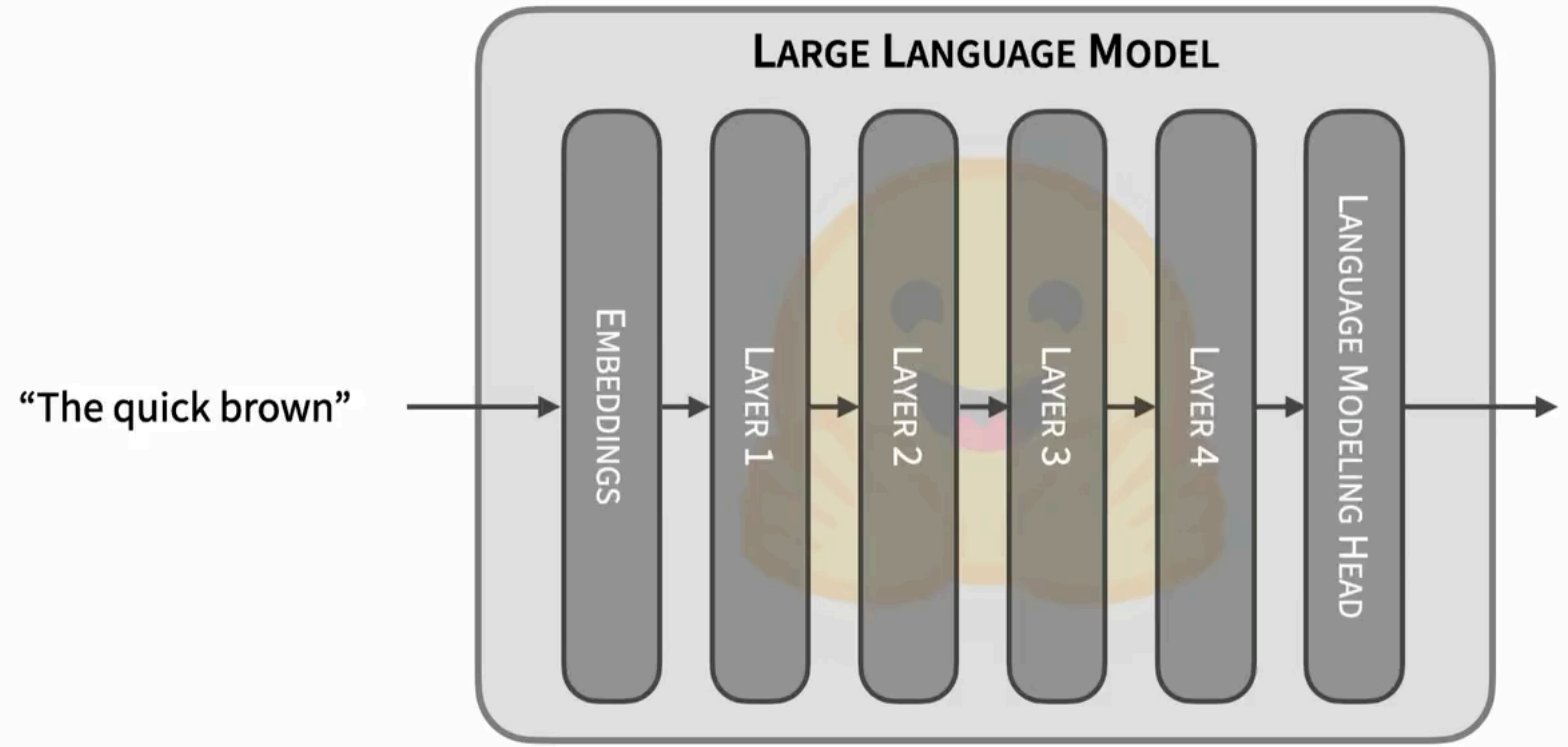
Causal Language Models



https://huggingface.co/docs/transformers/llm_tutorial

Causal Language Models

"Autoregressive generation iteratively selects the next token from a probability distribution to generate text"

Causal LMs: Common Pitfalls

- **Generated output is too short/long**: LM may require further tuning, also asking for more tokens can help

- **Incorrect generation mode**: greedy decoding or sampling? Which is better depends on your task

- **Wrong padding side**: you may need to pad the prompt text on the left to ensure that the input is the same size as the training phase of the LM.

- **Wrong prompt**: this is tricky and has produced a whole industry of "prompt engineering"

https://huggingface.co/docs/transformers/llm_tutorial

c.f. for code samples

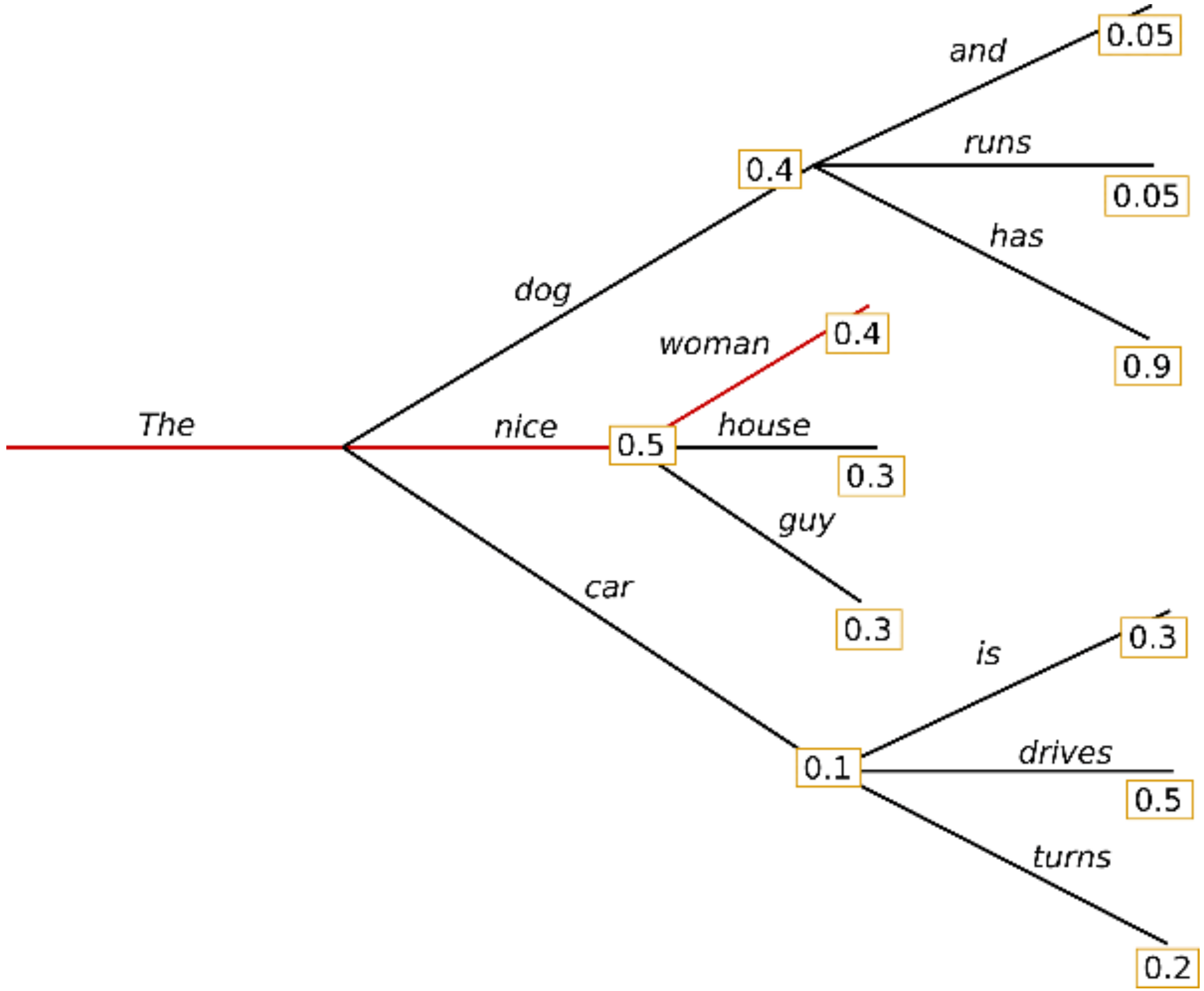
Decoding methods

https://huggingface.co/blog/how-to-generate

$$P(w_{1:T}|W_0) = \prod_{t=1}^{T} P(w_t|w_{1:t-1}, W_0) \text{ ,with } w_{1:0} = \emptyset,$$

- W_0 is the initial context word sequence (aka the "prompt")
- The length T of the word sequence is determined on-the-fly
- T is determined by the generation of the end-of-sentence EOS also known as the `<|endoftext|>` token
- The EOS token is produced like the other tokens from $P(w_t \mid w_{1:t-1}, W_0)$

Greedy Decoding



("The","nice","woman")
having an overall probability of

$$0.5 \times 0.4 = 0.2$$

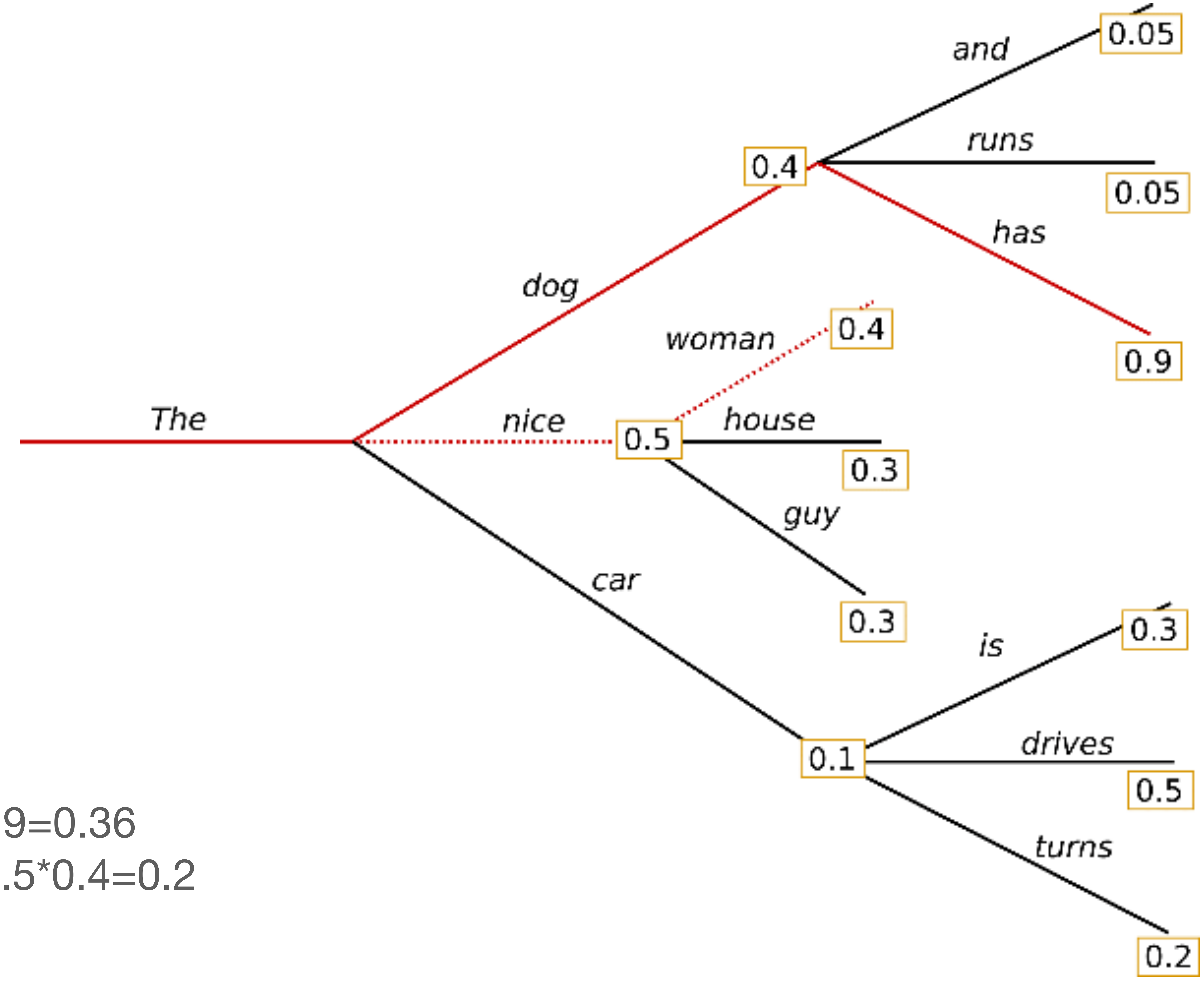
Beam Search

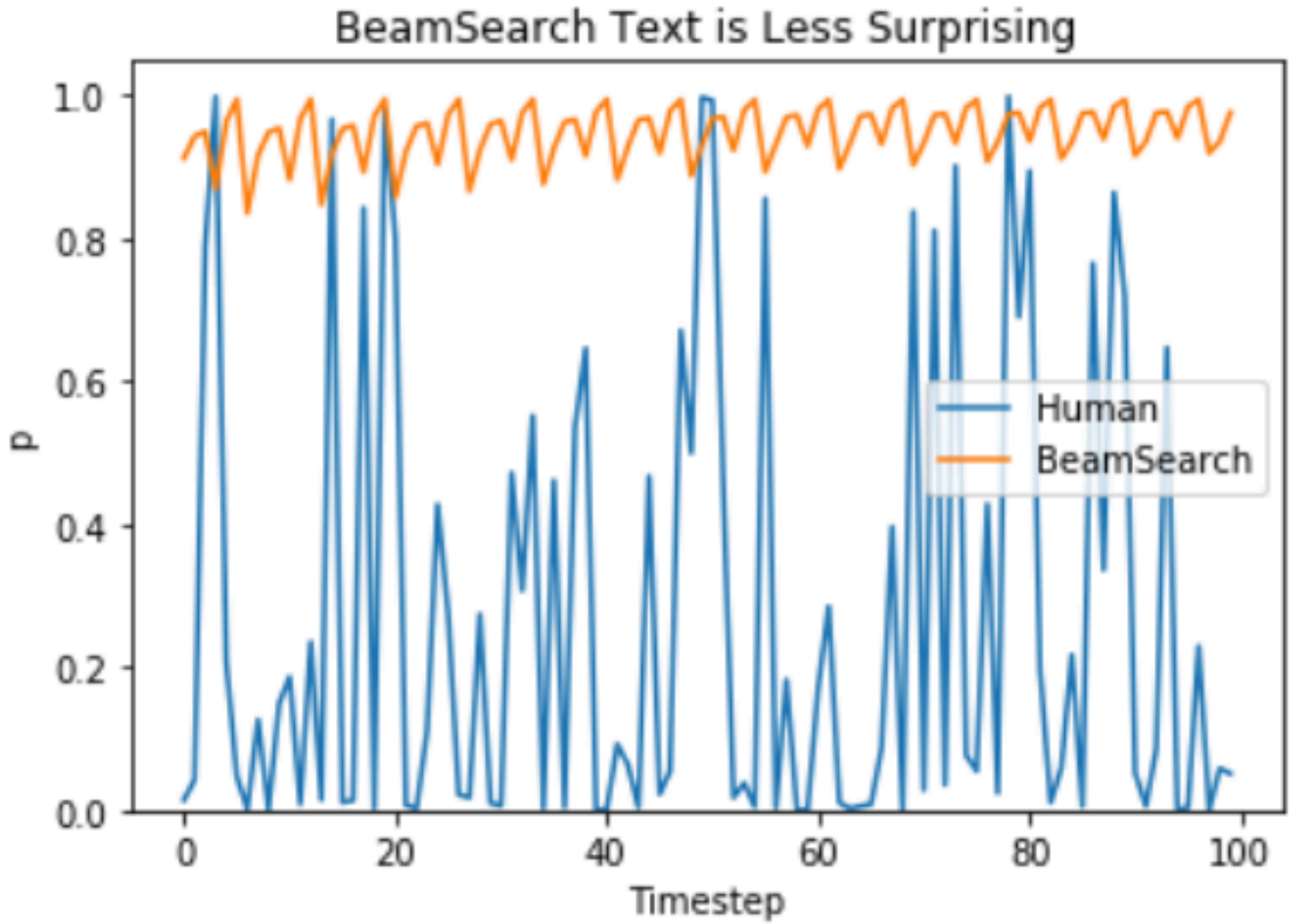
Let us assume a beam size of 2

Keep the 2 best outcomes at each time step

In this example:
("The", "nice") 0.5
("The", "dog") 0.4

Next time step:
("The", "dog", "has") 0.4*0.9=0.36
("The", "nice", "woman") 0.5*0.4=0.2





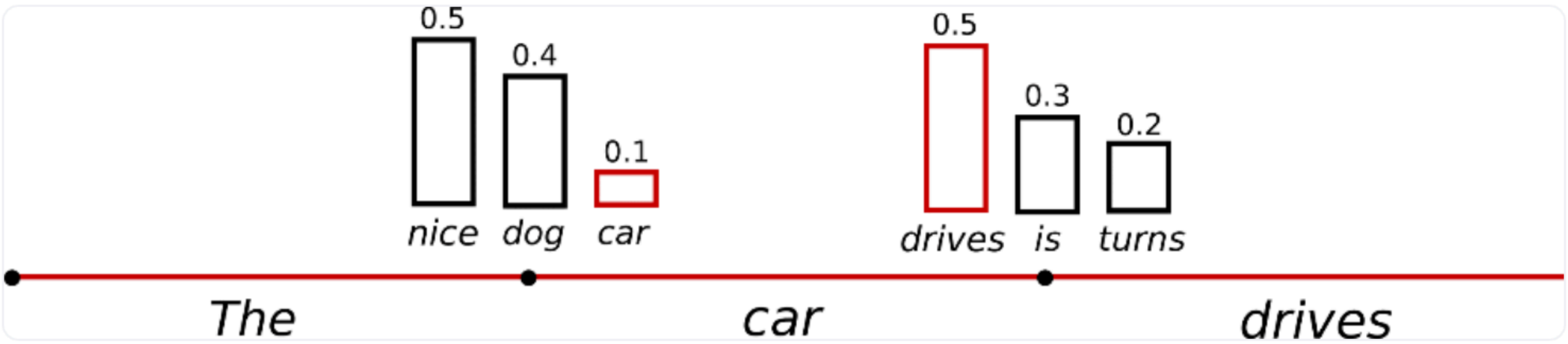
Ari Holtzman et al. (2019) plot probability that a model gives versus an estimate of the probability that a human would give. As humans we want generated text to surprise us and not be boring/predictable (depends on the task).

Beam Search Pitfalls

- Beam search can still be very repetitive.
  - Heuristic is to penalize repeated n-grams in the output.
  - Manually set the probability of next words that could create an already seen n-gram to 0
  - n should be greater than 2 or 3
- The choices in beam search may not be very diverse.
  - Similar continuations can happen due to common sub-trees in different branches
- These issues are referred to as **model degeneration**

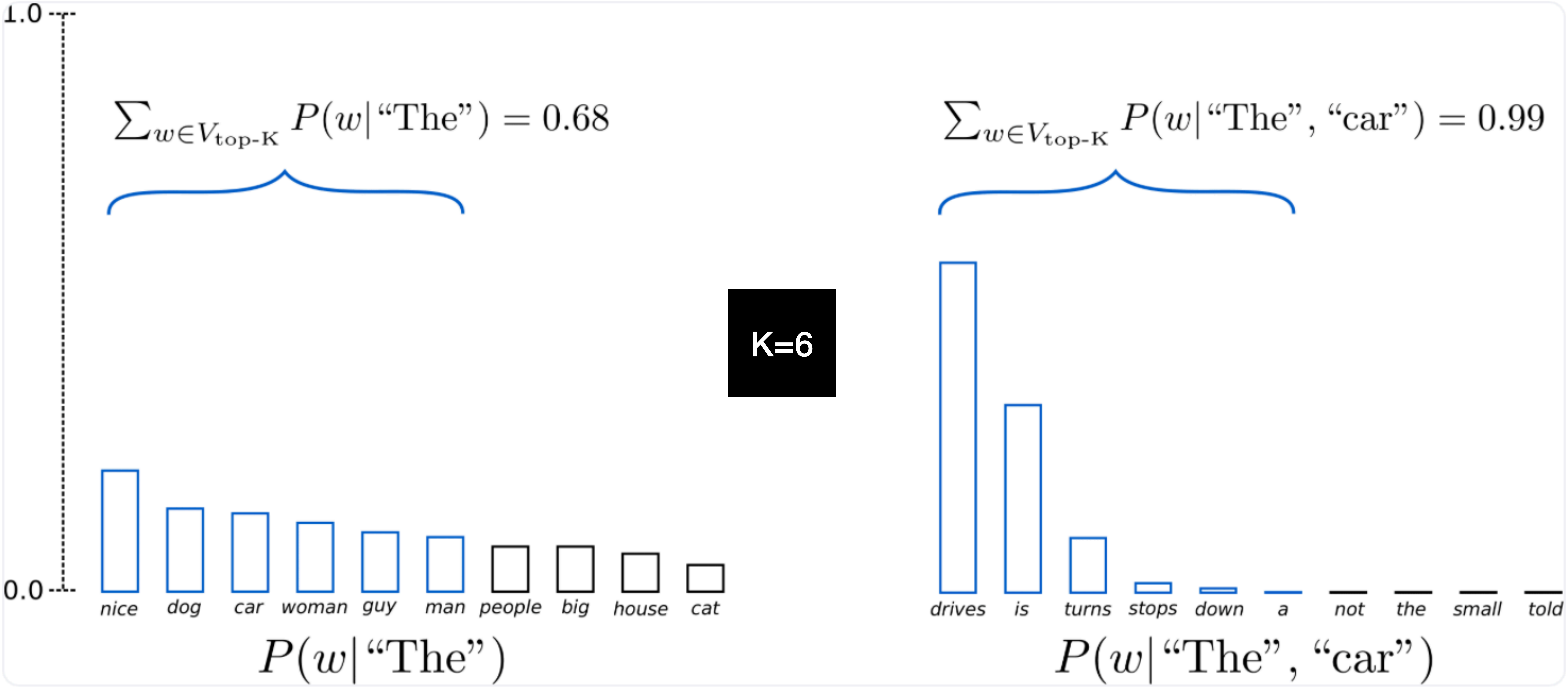
Sampling

- Sampling is represented by the operator $\sim$
- We pick the next word $w_t \sim P(w \mid w_{1:t-1}) = \dfrac{exp(logits(w \mid w_{1:t-1}))}{\sum_{w'} exp(logits(w' \mid w_{1:t-1}))}$
- Generation is no longer *deterministic.*
- Sampling can generate gibberish. Solution: use temperature $\dfrac{exp(logits(w \mid w_{1:t-1})/T)}{\sum_{w'} exp(logits(w' \mid w_{1:t-1})/T)}$



Top-k Sampling

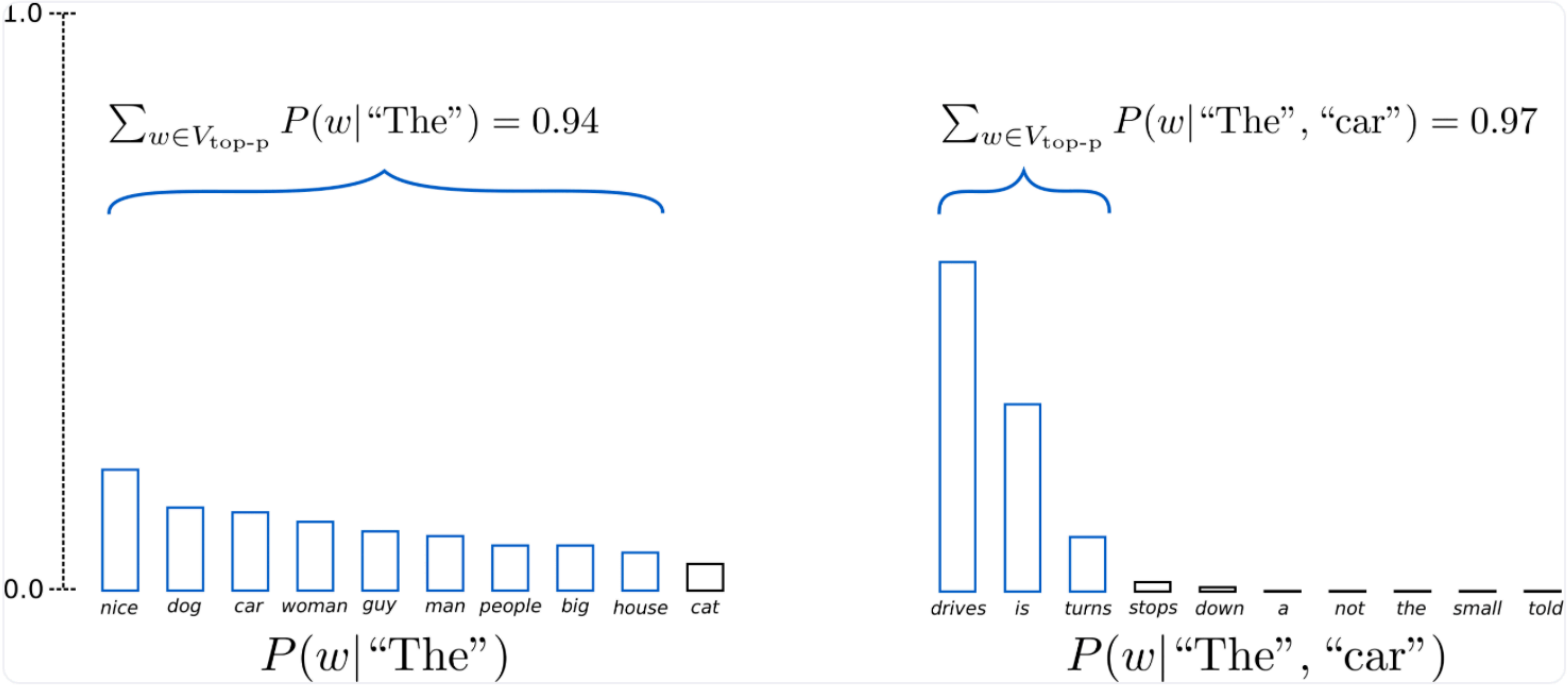
- K most likely next words are filtered and we re-normalize over the K words
- GPT2 showed that this worked better than beam search



Top-p Nucleus Sampling

- Choose the smallest set of words whose cumulative probability exceeds a threshold probability p. The probability mass is redistributed among this set of words.

- The size of the set being sampled from grows and shrinks depending on the probability distribution.

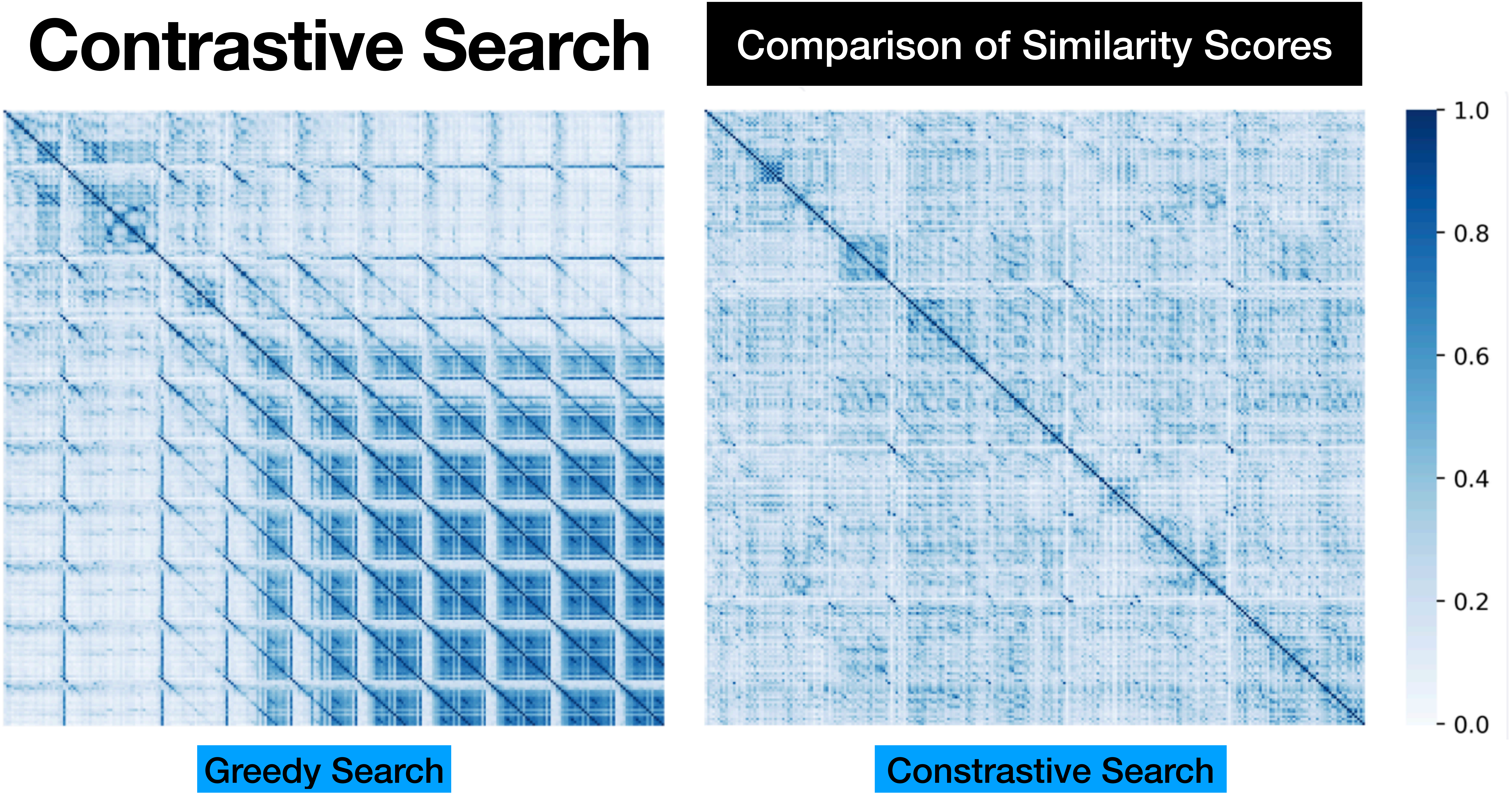


Contrastive Search

- Given a prefix text $\mathbf{x}_{<t}$ select the output next token x_t
- $V^{(k)}$ is the set of top-k predictions from the LM's probability distribution $p_\theta(v \mid \mathbf{x}_{<t})$ called the **model confidence**
- $\mathbf{s}(\cdot, \cdot)$ is the cosine similarity between two token representations is used to compute the **degeneration penalty**
- The more similar v is to the context the more we see **model degeneration.**
- Combine the two terms using a linear mixture.

$$x_t = \underset{v \in V^{(k)}}{\arg\max} \left\{ (1-\alpha) \times \underbrace{p_\theta(v|\boldsymbol{x}_{<t})}_{\text{model confidence}} - \alpha \times \underbrace{(\max\{s(h_v, h_{x_j}) : 1 \leq j \leq t-1\})}_{\text{degeneration penalty}} \right\},$$

Contrastive Search



Other problems

- **Unreachable subword problem**: there are some subwords for which under no circumstances is it possible to produce a subword (given any context).
- **Mode collapse**: tuning the LM might cause the model parameters to reach a state where Greedy and Sampling based generation produce the same output.
- **Softmax over very large vocabulary sizes**: Vocabulary sizes have reduced since subword segmentation has become the standard way to set up the vocabulary for LMs; However for very large vocabulary sizes, the compute efficiency for softmax might need careful consideration, e.g. use hierarchical softmax.

KV cache (and paged attention)

https://arxiv.org/pdf/2309.06180

Self-attention in auto-regressive LMs

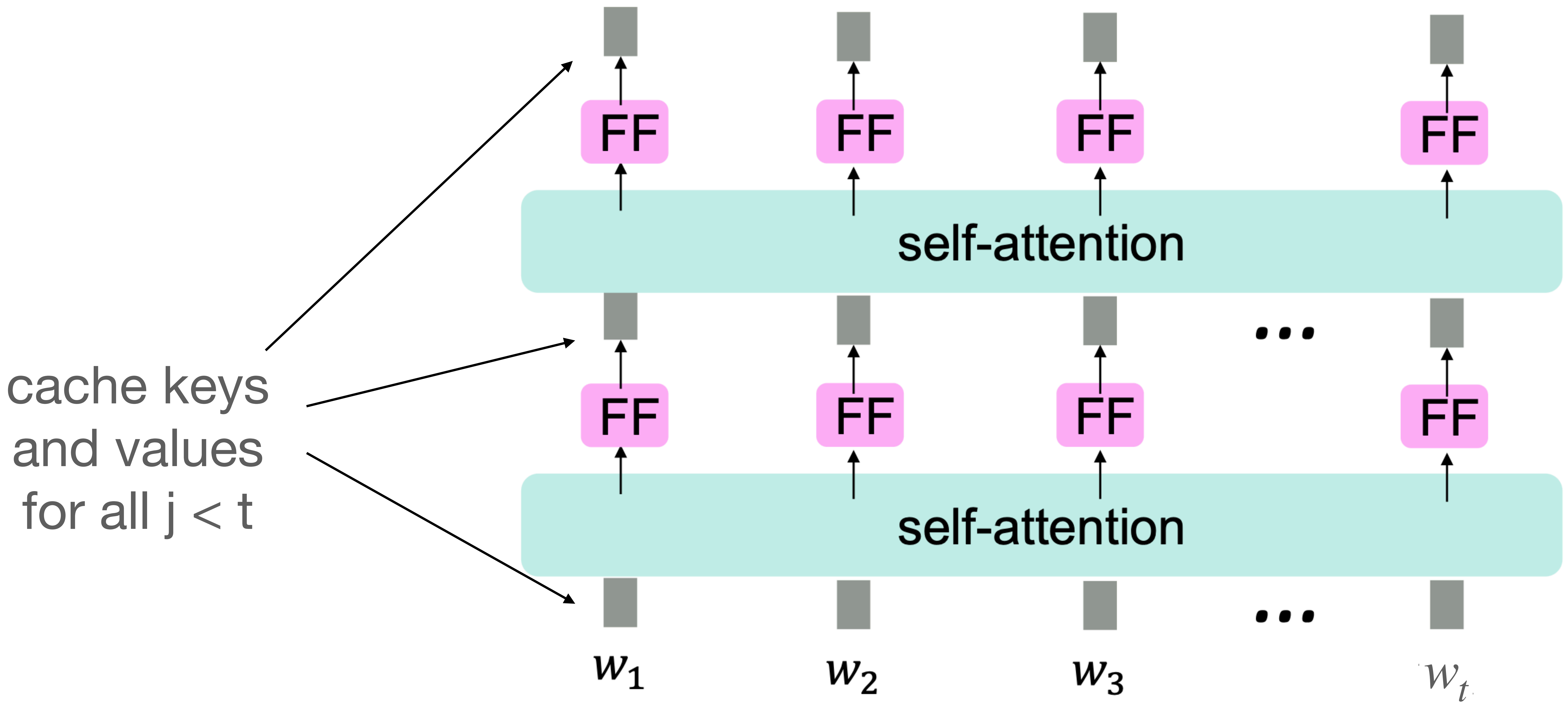
keys, queries and values computed from the history

- Let $\mathbf{w} = (w_1, \ldots, w_{t-1})$ be a sequence of tokens, like "Havana is the capital of"
- For each w_i let $\mathbf{x}_i = E\mathbf{w}_i$ where $E \in \mathbb{R}^{d \times |V|}$ is an embedding matrix. V is the vocabulary.
- Let Q, K, V be matrices in $\mathbb{R}^{d \times d}$
  - $\mathbf{q}_i = Q\mathbf{x}_i$
  - $\mathbf{k}_i = K\mathbf{x}_i$
  - $\mathbf{v}_i = V\mathbf{x}_i$

Attention layer output for next token is a weighted sum of values:

$$\mathbf{o}_t = \sum_{j<t} \mathit{softmax}_j(\mathbf{q}_t^T \mathbf{k}_j) \cdot \mathbf{v}_j$$

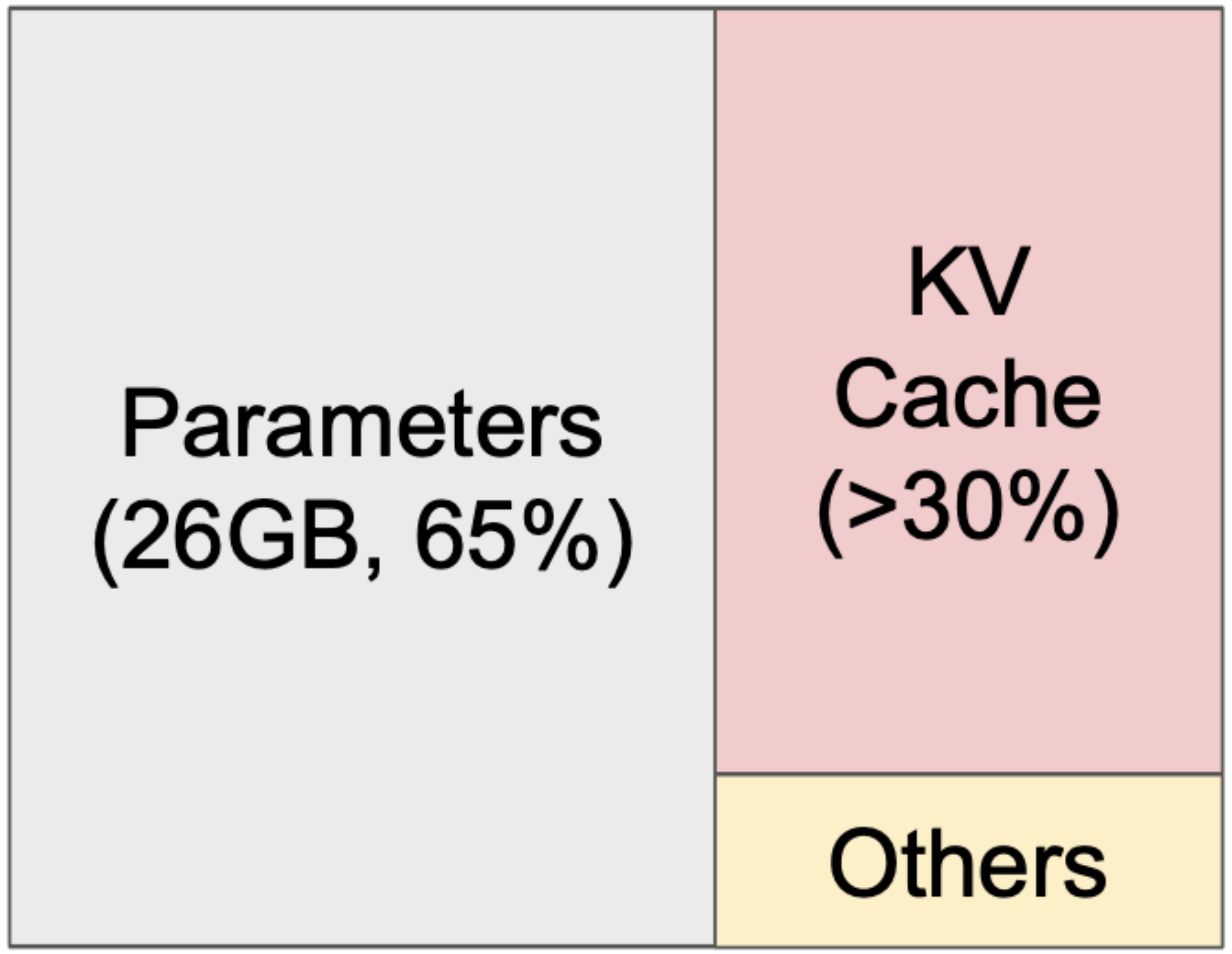
next token   previous tokens



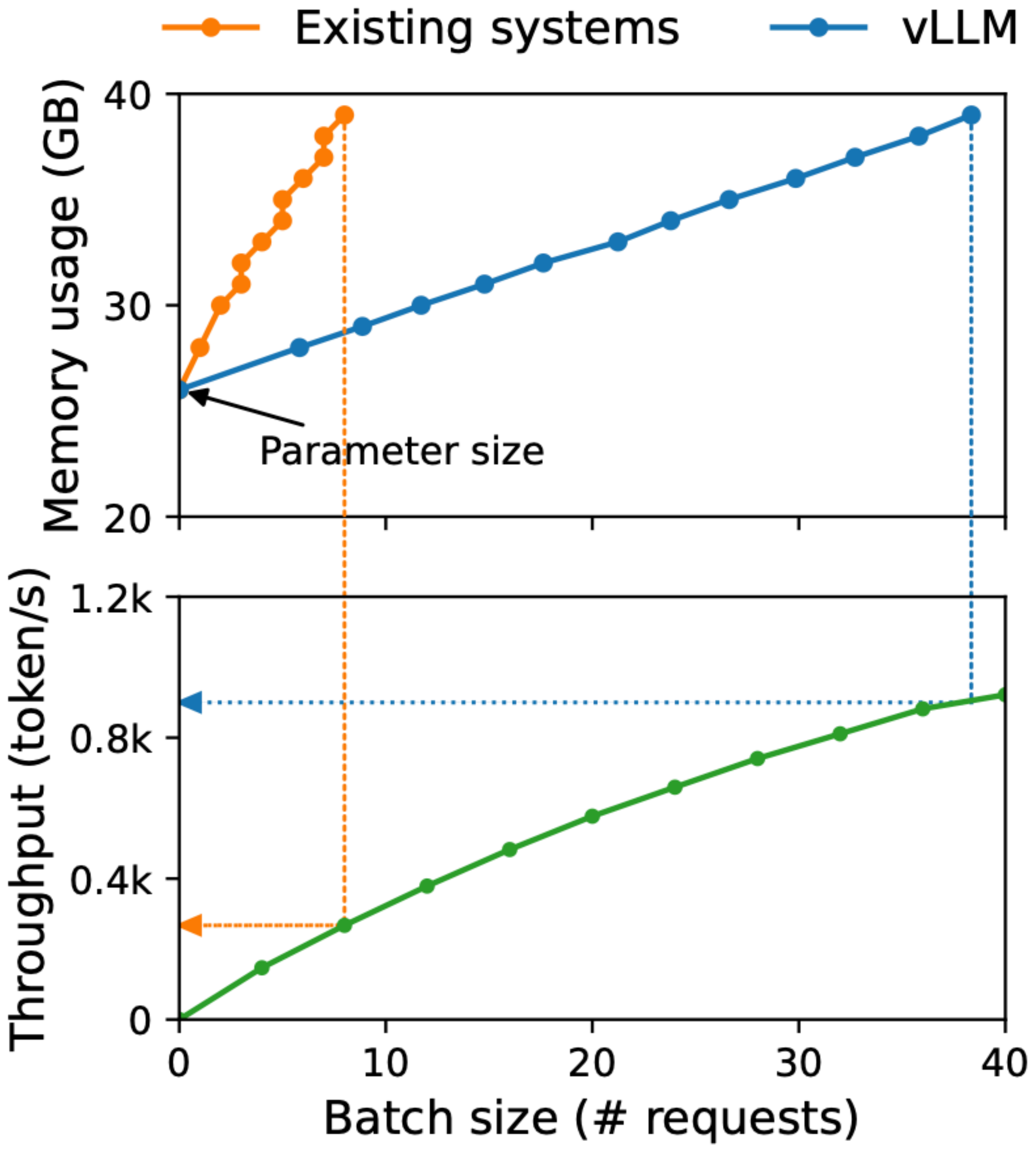
$$\mathbf{o}_t = \sum_{j<t} softmax_j(\mathbf{q}_t^T \mathbf{k}_j) \cdot \mathbf{v}_j$$

Phases of generation

- Prompt phase
    - all tokens are known
    - generate and store all the key and value tensors
- Autoregressive phase
    - compute the probability of the next token using cached tensors
    - add the key and value tensor to the cache



NVIDIA A100 40GB



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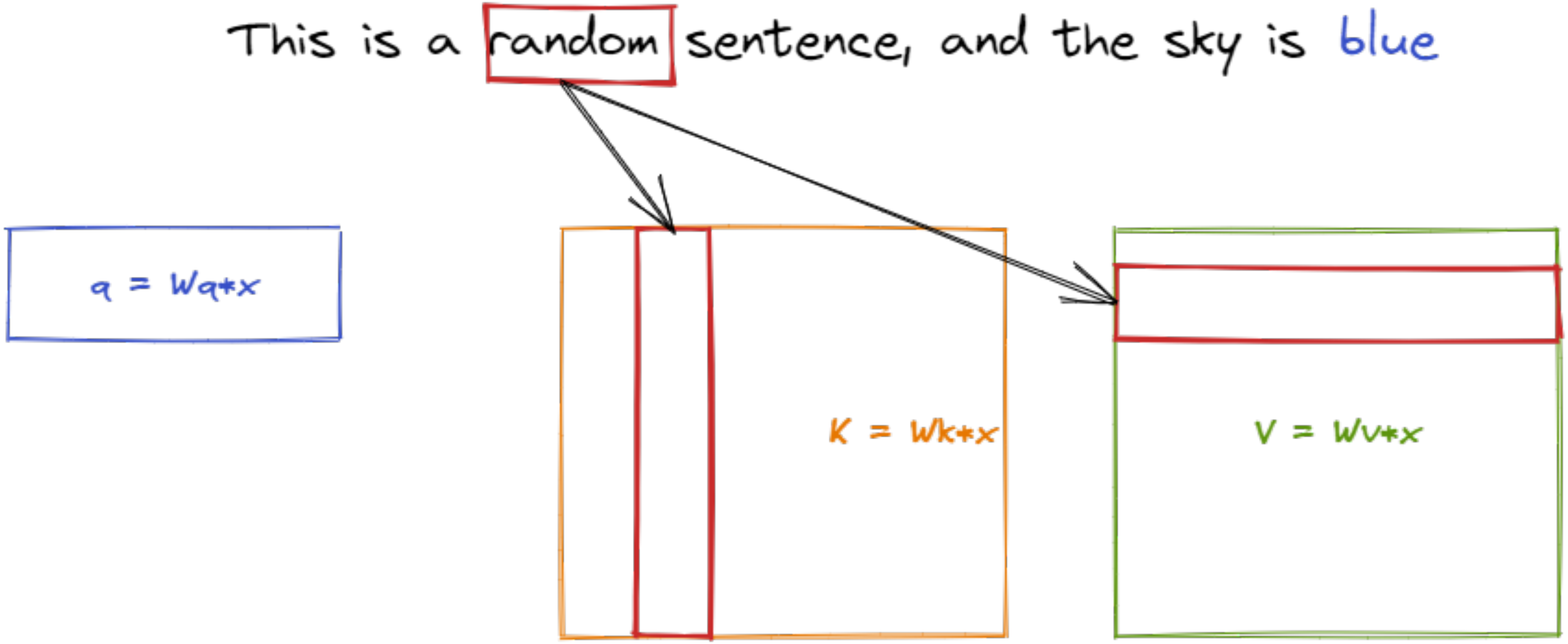
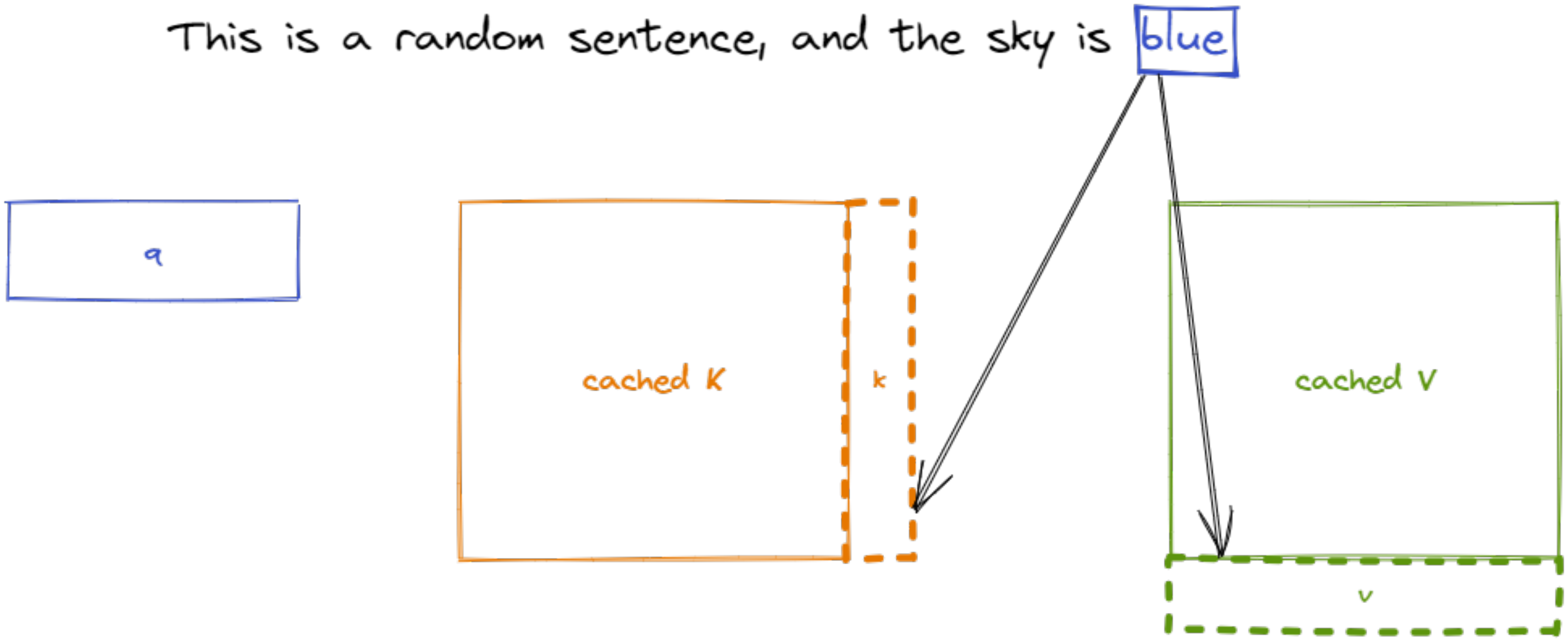


Naively, for each new token (e.g. "blue" above) the key and value vectors are recomputed every time, for each new token.

K and V contain information about the entire sequence, and query is the new token being added. So we are doing *softmax*(qK)V to compute attention.

https://mett29.github.io/posts/kv-cache/



During the sequence generation one token at a time, the two matrices and do not change very much

Once the embedding for the new token is computed, it's not going to change, no matter how many more tokens we generate

That is why the key and value vectors of existing tokens are often cached for generating future tokens. This approach leads to what is called the **KV cache**.

https://mett29.github.io/posts/kv-cache/

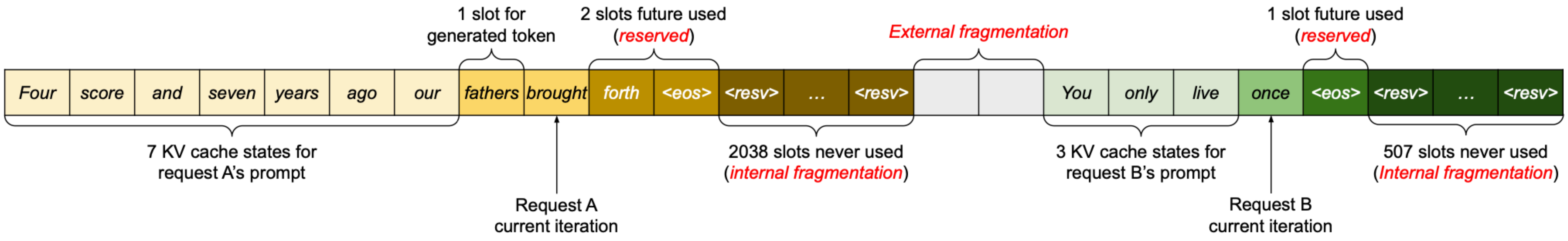


Figure 3. KV cache memory management in existing systems. Three types of memory wastes – reserved, internal fragmentation, and external fragmentation – exist that prevent other requests from fitting into the memory. The token in each memory slot represents its KV cache. Note the same tokens can have different KV cache when at different positions.

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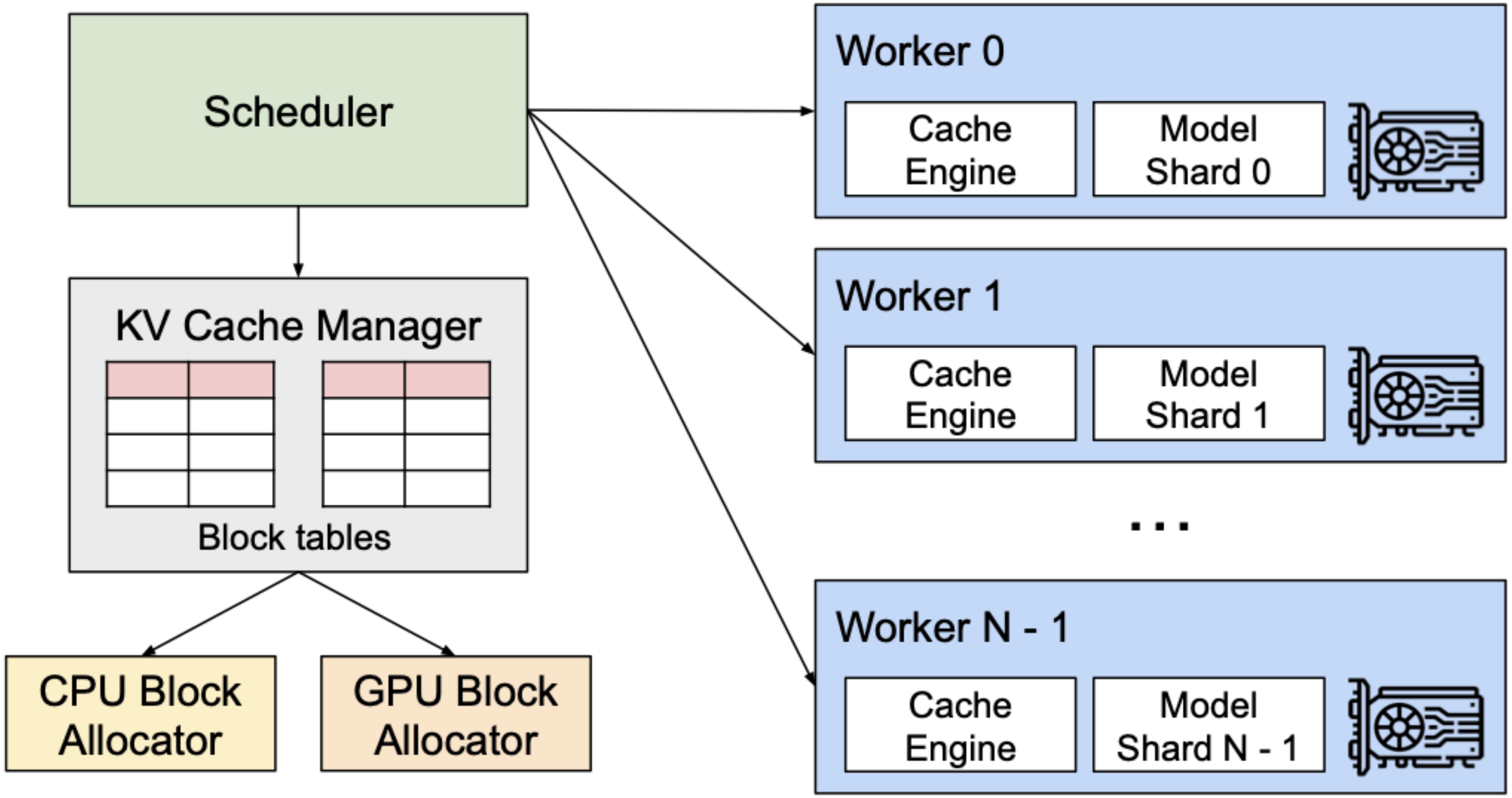
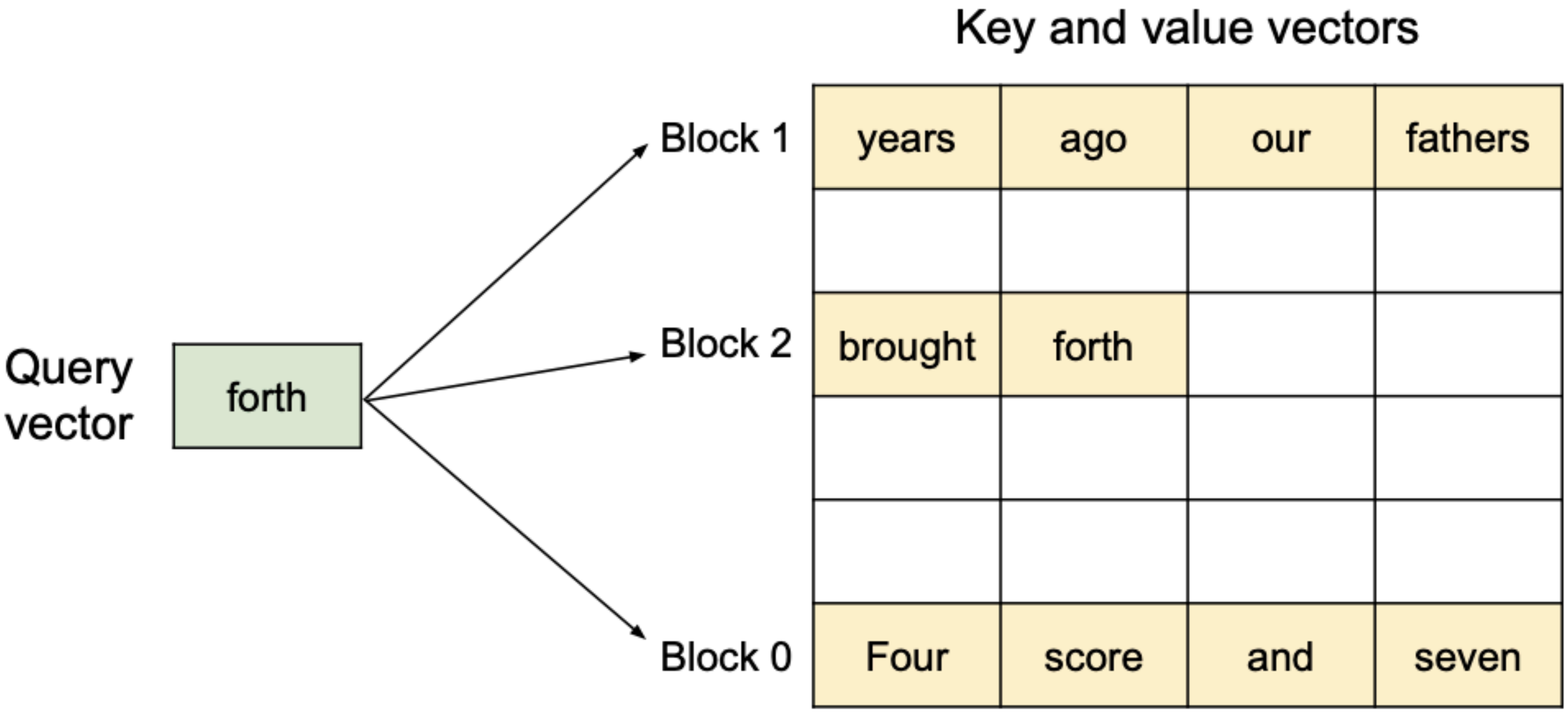


Figure 4. vLLM system overview.

https://arxiv.org/pdf/2309.06180

Paged Attention for efficient memory usage

implemented in the vllm inference engine



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